



AWS CloudTrail

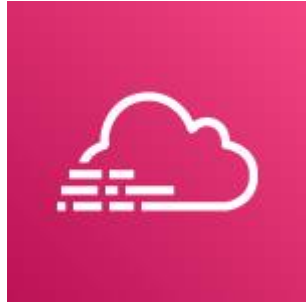
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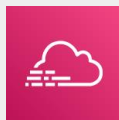
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Good luck on your exam!

CloudTrail



**Logs API calls between AWS services.
When you need to know **who to blame**.**



Introduction to CloudTrail

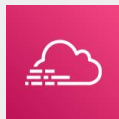
AWS CloudTrail is a service that enables **governance, compliance, operational auditing,** and **risk auditing** of your AWS account.

AWS CloudTrail is used to **monitor API calls** and **Actions** made on an AWS account.

Easily identify which users and accounts made the call to AWS eg.

- **Where** Source IP Address
- **When** EventTime
- **Who** User, UserAgent
- **What** Region, Resource, Action

```
1 {"Records": [{
2   "eventVersion": "1.0",
3   "userIdentity": {
4     "type": "IAMUser",
5     "principalId": "EX_PRINCIPAL_ID",
6     "arn": "arn:aws:iam::123456789012:user/Worf",
7     "accountId": "123456789012",
8     "accessKeyId": "EXAMPLE_KEY_ID",
9     "userName": "Worf"
10  },
11  "eventTime": "2014-03-24T21:11:59Z",
12  "eventSource": "iam.amazonaws.com",
13  "eventName": "CreateUser",
14  "awsRegion": "us-east-1",
15  "sourceIPAddress": "127.0.0.1",
16  "userAgent": "aws-cli/1.3.2 Python/2.7.5 Windows/10",
17  "requestParameters": {"userName": "LaForge"},
18  "responseElements": {"user": {
19    "createDate": "Mar 24, 2014 9:11:59 PM",
20    "userName": "LaForge",
21    "arn": "arn:aws:iam::123456789012:user/LaForge",
22    "path": "/",
23    "userId": "EXAMPLEUSERID"
24  }}
25 ]}]}
```



CloudTrail – Event History

CloudTrail is already logging by default and will collect logs for **last 90 days** via **Event History**

If you need more than 90 days you need to create a **Trail**

Trails are output to S3 and do not have GUI like Event History. To analyze a Trail you'd have to use  **Amazon Athena**.

CloudTrail

Dashboard

Event history

Trails

Learn more

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Event history

Your event history contains the activities taken by people, groups, or AWS services in [supported services](#). It filters out read-only events. You can change or remove that filter, or apply other filters.

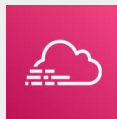
You can view the last 90 days of events. Choose an event to view more information about it. To view a console trail and then go to your Amazon S3 bucket or CloudWatch Logs. [Learn more](#)

Can't find what you're looking for? [Run advanced queries in Amazon Athena](#)

Filter: Read only false ⊗

Time range: Select time range

	Event time	User name	Event name
▶	2019-09-01, 09:33:07 PM	i-014d0d0e482491e69	UpdateInstanceInformation
▶	2019-09-01, 09:30:07 PM	i-08ece9e263d3edfbc	UpdateInstanceInformation
▶	2019-09-01, 09:28:07 PM	i-0984241e0f6a0f9ca	UpdateInstanceInformation
▶	2019-09-01, 09:25:07 PM	i-07a9e824ebb4d5f2b	UpdateInstanceInformation
▶	2019-09-01, 09:23:34 PM	exampro-events	CreateLogStream
▶	2019-09-01, 09:23:07 PM	i-014d0d0e482491e69	UpdateInstanceInformation
▶	2019-09-01, 09:20:07 PM	i-0f5f9d47f3c1cfe6d	UpdateInstanceInformation
▶	2019-09-01, 09:18:07 PM	i-08ece9e263d3edfbc	UpdateInstanceInformation
▶	2019-09-01, 09:15:07 PM	i-07a9e824ebb4d5f2b	UpdateInstanceInformation
▶	2019-09-01, 09:13:51 PM	exampro-metrics	CreateLogStream



CloudTrail – Trail Options

A Trail can be set to **log to all regions**

Apply trail to all regions ☒ Yes ☐ No

Creates the same trail in all regions and delivers log files for all regions

A Trail can be set to **across all accounts in an Organization**

Apply trail to my organization ☒ Yes ☐ No 

You can **Encrypt your Logs** using Server Side Encryption via Key Management Service (SSE-KMS)

Encrypt log files with SSE-KMS ☒ Yes ☐ No 

Create a new KMS key ☒ Yes ☐ No

We can ensure the Integrity of our logs to see if they have been tampered we need to turn on **Log File Validation**

Create a new S3 bucket ☒ Yes ☐ No

S3 bucket* cloudtrail 

▼ Advanced

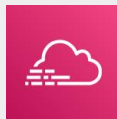
Log file prefix 

Location: /AWSLogs/655604346524/CloudTrail/us-east-1

Encrypt log files with SSE-KMS ☐ Yes ☒ No 

Enable log file validation ☒ Yes ☐ No 

Send SNS notification for every log file delivery ☐ Yes ☒ No 



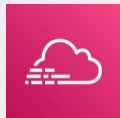
CloudTrail to CloudWatch

CloudTrail can be set to deliver events to a CloudWatch log.

▼ CloudWatch Logs

Configuring delivery to CloudWatch Logs enables you to receive SNS notifications from CloudWatch when specific API activity occurs. Standard CloudWatch and CloudWatch Logs [charges](#) will apply. [Learn more](#)

Configure



CloudTrail – Management Events vs Data Events

Management Events

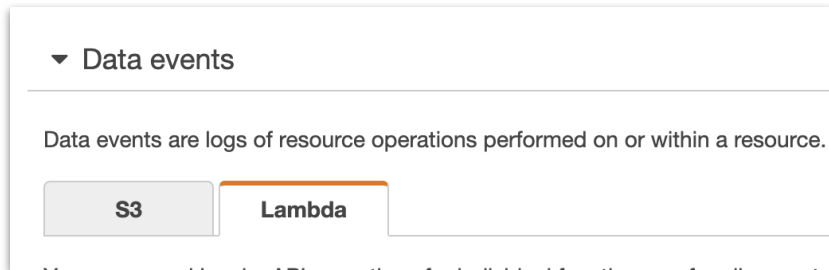
Tracks management operations. Turned on by default. Can't be turned off.

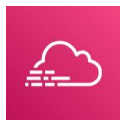
- **Configuring security**
eg. IAM AttachRolePolicy API operations
- **Registering devices**
eg. Amazon EC2 CreateDefaultVpc API operations)
- **Configuring rules for routing data**
eg. Amazon EC2 CreateSubnet API operations
- **Setting up logging**
eg. AWS CloudTrail CreateTrail API operations

Data Events

Tracks specific operations for specific AWS Services. Data events are high volume logging and will result in additional charges. **Turned off by default**

The two services that can be tracked is S3 and Lambda. So it would track action such as: GetObject, DeleteObject, PutObject





CloudTrail *CheatSheet*

- CloudTrail logs calls between AWS services
- **governance, compliance, operational auditing**, and **risk auditing** are keywords relating to CloudTrail
- When you need to know **who to blame** *think* CloudTrail
- CloudTrail by default logs event data for the past 90s days via **Event History**
- To track beyond 90 days you need to create **Trail**
- To ensure logs have not been tampered with you need to turn on **Log File Validation** option
- CloudTrail logs can be encrypted using **KMS (Key Management Service)**
- CloudTrail can be set to log across all AWS accounts in an Organization and all regions in an account.
- CloudTrail logs can be streamed to CloudWatch logs
- Trails are outputted to an S3 bucket that you specify
- CloudTrail logs two kinds of events: **Management Events** and **Data Events**
- **Management events** log management operations eg. AttachRolePolicy
- **Data Events** log data operations for resources (S3, Lambda) eg. GetObject, DeleteObject, and PutObject
- Data Events are **disabled** by default when creating a Trail.
- Trail which are logged within S3 can be analyzed using Athena